



KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

SECTION A [25 MARKS]

This section consists of **3 questions**. Answer **all** questions.

1. Solve

(a) If $f(x) = x^2 + 3$. Find $\lim_{x \rightarrow 0} \left(\frac{f(x+5) - f(5)}{x} \right)$

[5 marks]

(b) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 3} - x)$

[4 marks]

2. (a) Given that f and g are functions such that $f(x) = x^3 g(x^2)$, $g(1) = 5$ and $g'(1) = 3$.
Find the value of $f'(1)$.

[5 marks]

(b) Given $f(x) = \frac{3x^2 - 5}{x}$ and $g(x) = \ln \sqrt{f(x)}$, find $g'(x)$.

[4 marks]

3. Given the curve $y = x^3 - \frac{3}{2}x^2 - 18x + 1$. Use the first derivative test to determine the maximum and minimum points of y .

[7 marks]





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

SECTION B [75 MARKS]

1. Given a complex number $z = \frac{(1+i)^2}{4-i}$.

(a) Express z in the cartesian form $a+bi$, where a and b are real numbers. **[4 marks]**

(b) Find the argument of z correct to 3 significant numbers. **[2 marks]**

2. Find

(a) the value of x that satisfies the equation $\log_2 x = 2\log_4(x+4) - 1$. **[5 marks]**

(b) the solution set for the inequality $9x + \frac{25}{x} \geq 30$ **[5 marks]**

3. Given that $f(x) = \sqrt{x+4}$ where $x \geq -4$.

(a) Show that the function is a one-to-one function algebraically. **[4 marks]**

(b) Find $f^{-1}(x)$. **[3 marks]**

(c) Sketch graphs of $f(x)$ and $f^{-1}(x)$ on the same coordinate axes. **[3 marks]**

4. (a) If $g(x) = \frac{2x-3}{mx+2}$ where $m \neq 0$ and $x \neq -\frac{2}{m}$ such that $(g \circ g)(0) = \frac{12}{5}$.

Calculate the value of m .

[4 marks]

(b) By finding the composite functions $(f \circ g)(x)$ and $(g \circ f)(x)$, determine if

$f(x) = \frac{1}{3x+2}$ and $g(x) = \frac{1-2x}{3x}$ are inverse function of each other.

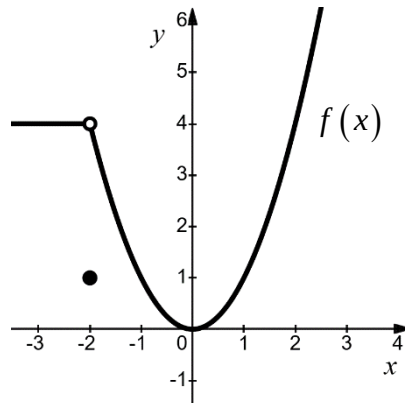
[6 marks]





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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

5. a)



From the graph of $f(x)$ above, determine ;

i) $\lim_{x \rightarrow -2} f(x)$, if exist.

[2marks]

ii) whether $f(x)$ is continuous at $x = -2$.

[2marks]

b) Given
$$f(x) = \begin{cases} x+1 & x < -1 \\ x^2 - 2x - m & -1 \leq x < 0 \\ \frac{3x+1}{\sqrt{x^2-1}} & x \geq 0 \end{cases}$$

i) Find the value of m given that $f(x)$ is continuous at $x = -1$.

[4marks]

ii) Find the vertical and horizontal asymptote of $f(x)$ for $x \geq 0$, if exist. **[6marks]**





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

6. a) If $y = \sin(x^2 + 3)$, show that $x \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3y = 0$. **[5 marks]**

b) The parametric equations of a curve are given by $x = \cot(t) - \operatorname{cosec}(t)$ and $y = \tan(t) - \sec(t)$ where $0 < t < \frac{1}{2}\pi$.

i) Show that $x - \frac{1}{x} = 2 \cot(t)$ and $y - \frac{1}{y} = 2 \tan(t)$.

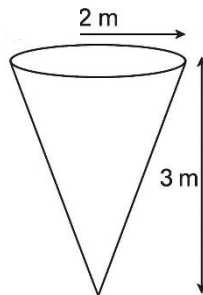
Deduce the value of $\left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right)$. **[5marks]**

ii) Find $\frac{dy}{dx}$ and express your answer in terms of x and y only.

Hence, determine the gradient of the curve at the point $\left(-\frac{1}{2}, -\frac{1}{3}\right)$.

[6marks]

7. An inverted conical tank with radius $2m$ and height $3m$ is completely filled with water. Water is leaking from the vertex of the tank at a rate of $0.1m^3$ per minute.



At the instant when the height of the water is $1m$, calculate the rate of change of

a) the water level. **[7marks]**

b) the radius of the water surface. **[2marks]**





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SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

ANSWERS SCHEME

SECTION A (25 MARKS)

1. Solve

(a) If $f(x) = x^2 + 3$. Find $\lim_{x \rightarrow 0} \left(\frac{f(x+5) - f(5)}{x} \right)$

[5 marks]

(b) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 3} - x)$

[4 marks]

NO	ANSWER SCHEME	MARKS
1	(a) $f(x) = x^2 + 3$ $f(x+5) = (x+5)^2 + 3$ $= x^2 + 10x + 25 + 3$ $= x^2 + 10x + 28$ $f(5) = 5^2 + 3 = 28$ $\lim_{x \rightarrow 0} \left(\frac{f(x+5) - f(5)}{x} \right)$ $= \lim_{x \rightarrow 0} \frac{x^2 + 10x + 28 - 28}{x}$ $= \lim_{x \rightarrow 0} \frac{x^2 + 10x}{x}$ $= \lim_{x \rightarrow 0} \frac{x(x+10)}{x}$ $= \lim_{x \rightarrow 0} (x+10)$ $= 10$ (b) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 3} - x)$ $= \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 3} - x) \times \left(\frac{\sqrt{x^2 + 3} + x}{\sqrt{x^2 + 3} + x} \right)$ $= \lim_{x \rightarrow +\infty} \left(\frac{3}{\sqrt{x^2 + 3} + x} \right)$ $= \frac{3}{\infty}$ $= 0$	K1✓ K1 (substitute value) K1✓ (simplify) K1(factorize) J1 B1 (conjugate) K1✓ (simplify) J1





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

	Total	9 marks

2. (a) Given that f and g are functions such that $f(x) = x^3 g(x^2)$, $g(1) = 5$ and $g'(1) = 3$.

Find the value of $f'(1)$.

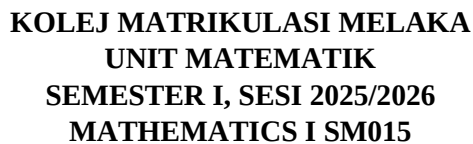
[5 marks]

(b) Given $f(x) = \frac{3x^2 - 5}{x}$ and $g(x) = \ln \sqrt{f(x)}$, find $g'(x)$.

[4 marks]

2	(a) $f(x) = x^3 g(x^2)$ $f'(x) = x^3 \frac{d}{dx} [g(x^2)] + g(x^2) \frac{d}{dx} (x^3)$ $= x^3 [g'(x^2)(2x)] + g(x^2) 3(x^2)$ $= 2x^4 g'(x^2) + 3(x^2) g(x^2)$ when $x = 1$, given $g(1) = 5$ and $g'(1) = 3$ $f'(1) = 2(1)^4 g'(1) + 3(1)^2 g(1)$ $= 2(3) + 3(5)$ $= 21$ (b) $g(x) = \ln \sqrt{\frac{3x^2 - 5}{x}}$ $= \ln \left(\frac{3x^2 - 5}{x} \right)^{\frac{1}{2}}$ $= \frac{1}{2} \ln \left(\frac{3x^2 - 5}{x} \right)$ $= \frac{1}{2} [\ln(3x^2 - 5) - \ln x]$ $g'(x) = \frac{1}{2} \left(\frac{6x}{3x^2 - 5} - \frac{1}{x} \right)$ $= \frac{1}{2} \left(\frac{6x^2 - 3x^2 + 5}{x(3x^2 - 5)} \right)$ $= \frac{1}{2} \left(\frac{3x^2 + 5}{x(3x^2 - 5)} \right)$	K1 K1√ (Simplify) J1 K1 (substitute value) J1 K1 K1 K1√ J1
	Total	9 marks







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MATHEMATICS I SM015

SECTION B (75 MARKS)

1. Given a complex number $z = \frac{(1+i)^2}{4-i}$.

(a) Express z in the cartesian form $a+bi$, where a and b are real numbers. [4 marks]

(b) Find the argument of z correct to 3 significant numbers. [2 marks]

NO.	ANSWER SCHEME	MARKS
1(a)	$z = \frac{(1+i)^2}{4-i}$ $= \frac{1+2i+i^2}{4-i}$ $= \frac{1+2i+(-1)}{4-i}$ $= \frac{2i}{4-i} \times \frac{4+i}{4+i}$ $= \frac{8i+2i^2}{16-i^2}$ $= \frac{8i+2(-1)}{16-(-1)}$ $= -\frac{2}{17} + \frac{8}{17}i$	K1-expand K1-multiply conjugate K1-sub $i^2 = -1$ J1
1(b)	$\theta = \pi - \tan^{-1} \left(\frac{\frac{8}{17}}{\frac{2}{17}} \right)$ $= 1.82 \text{ rad}$	K1-correct formula J1-must 3sf
	TOTAL MARKS :	6 marks





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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

2. Find

(a) the value of x that satisfies the equation $\log_2 x = 2\log_4(x+4) - 1$.

[5 marks]

(b) the solution set for the inequality $9x + \frac{25}{x} \geq 30$

[5 marks]

NO.	ANSWER SCHEME	MARKS
2(a)	$\log_2 x = 2\log_4(x+4) - 1$ $\log_2 x = 2\left(\frac{\log_2(x+4)}{2}\right) - \log_2 2$ $\log_2 x = \log_2(x+4) - \log_2 2$ $\log_2 x = \log_2 \frac{(x+4)}{2}$ $x = \frac{(x+4)}{2}$ $x = 4$	K1-change base K1-simplify K1-use rule of log K1 J1
2(b)	$9x + \frac{25}{x} \geq 30$ $9x + \frac{25}{x} - 30 \geq 0$ $\frac{9x^2 + 25 - 30x}{x} \geq 0$ $\frac{(3x-5)^2}{x} \geq 0$ Since $(3x-5)^2 \geq 0$, then $x > 0$. The solution set is $\{x : x > 0\}$.	K1- group to LHS K1- equalize denominator K1- factorize numerator K1J1
	TOTAL MARKS :	10 marks





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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

3. Given that $f(x) = \sqrt{x+4}$ where $x \geq -4$.

(a) Show that the function is a one-to-one function algebraically.

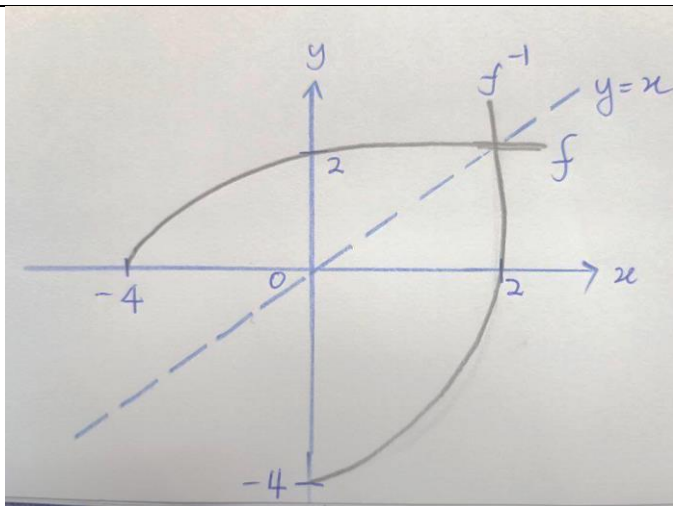
[4 marks]

(b) Find $f^{-1}(x)$.

[3 marks]

(c) Sketch graphs of $f(x)$ and $f^{-1}(x)$ on the same coordinate axes.

[3 marks]

NO.	ANSWER SCHEME	MARKS
3(a)	Let $x_1, x_2 \in D_f$: $f(x_1) = f(x_2)$ $\sqrt{x_1 + 4} = \sqrt{x_2 + 4}$ $(\sqrt{x_1 + 4})^2 = (\sqrt{x_2 + 4})^2$ $x_1 + 4 = x_2 + 4$ $x_1 = x_2$ Since $f(x_1) = f(x_2)$ implies $x_1 = x_2$, then $f(x) = \sqrt{x+4}$ is a one-to-one function.	K1-start with $f(x_1) = f(x_2)$ J1 K1 J1
3(b)	$\sqrt{f^{-1}(x) + 4} = x$ $f^{-1}(x) + 4 = x^2$ $f^{-1}(x) = x^2 - 4$	K1 K1 J1
3(c)		R1-shape R1-line $y = x$, intersection points R1-label graph f and f^{-1}
	TOTAL MARKS :	10 marks





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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

4. (a) If $g(x) = \frac{2x-3}{mx+2}$ where $m \neq 0$ and $x \neq -\frac{2}{m}$ such that $(g \circ g)(0) = \frac{12}{5}$.

Calculate the value of m .

[4 marks]

- (b) By finding the composite functions $(f \circ g)(x)$ and $(g \circ f)(x)$, determine if

$$f(x) = \frac{1}{3x+2} \text{ and } g(x) = \frac{1-2x}{3x} \text{ are inverse function of each other.}$$

[6 marks]

NO.	ANSWER SCHEME	MARKS
4(a)	$(g \circ g)(0) = \frac{12}{5}$ $g[g(0)] = \frac{12}{5}$ $g\left[\frac{2(0)-3}{m(0)+2}\right] = \frac{12}{5}$ $g\left(-\frac{3}{2}\right) = \frac{12}{5}$ $\frac{2\left(-\frac{3}{2}\right)-3}{m\left(-\frac{3}{2}\right)+2} = \frac{12}{5}$ $\frac{-6}{-3m+4} = \frac{12}{5}$ $\frac{-6}{-3m+4} = \frac{12}{5}$ $m = 3$	K1- sub 0 K1- get $-\frac{3}{2}$ K1-attempt to find m J1
4(b)	$(f \circ g)(x) = f[g(x)]$ $= f\left[\frac{1-2x}{3x}\right]$ $= \frac{1}{3\left(\frac{1-2x}{3x}\right)+2}$ $= \frac{1}{1-2x+2x}$ $= \frac{1}{x}$ $= \frac{1}{x}$ $= x$	K1-correct sub in composite functions J1- get x





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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

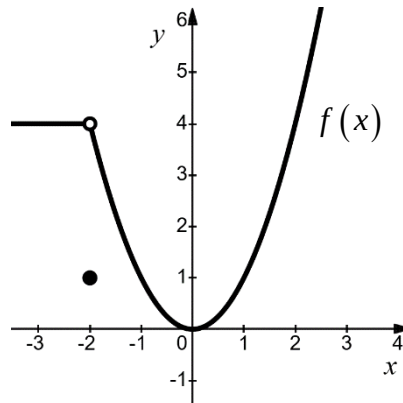
	$(g \circ f)(x) = g[f(x)]$ $= g\left[\frac{1}{3x+2}\right]$ $= \frac{1 - 2\left(\frac{1}{3x+2}\right)}{3\left(\frac{1}{3x+2}\right)}$ $= \frac{3x+2-2}{3}$ $= \frac{3x}{3}$ $= x$ Since $(f \circ g)(x) = (g \circ f)(x) = x$, then $f(x)$ and $g(x)$ are inverse of each other.	K1-correct sub in composite functions J1- get x K1 J1
	TOTAL MARKS :	10 marks





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

5. a)



From the graph of $f(x)$ above, determine ;

i) $\lim_{x \rightarrow -2} f(x)$, if exist.

[2marks]

ii) whether $f(x)$ is continuous at $x = -2$.

[2marks]

b) Given
$$f(x) = \begin{cases} x+1 & x < -1 \\ x^2 - 2x - m & -1 \leq x < 0 \\ \frac{3x+1}{\sqrt{x^2-1}} & x \geq 0 \end{cases}$$

i) Find the value of m given that $f(x)$ is continuous at $x = -1$.

[4marks]

ii) Find the vertical and horizontal asymptote of $f(x)$ for $x \geq 0$, if exist. **[6marks]**





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

NO	ANSWER SCHEME	MARKS
5 a) i)	$\lim_{x \rightarrow -2^-} f(x) = 4 \quad \lim_{x \rightarrow -2^+} f(x) = 4$ Since $\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^+} f(x)$, $\lim_{x \rightarrow -2} f(x)$ exist. $\lim_{x \rightarrow -2} f(x) = 4$	K1 – Finding $\lim_{x \rightarrow -2^+} f(x)$ and $\lim_{x \rightarrow -2^-} f(x)$ J1
5 a) ii)	$f(-2) = 1$ $\lim_{x \rightarrow -2} f(x) = 4$ Since $f(-2) \neq \lim_{x \rightarrow -2} f(x)$, $f(x)$ is not continuous at $x = -2$.	K1 – identify $f(-2)$. J1
5 b) i)	continuous at $x = -1$ $f(-1) = \lim_{x \rightarrow -1} f(x)$ $f(-1) = \lim_{x \rightarrow -1^-} f(x)$ $(-1)^2 - 2(-1) - m = \lim_{x \rightarrow -1^-} x + 1$ $3 - m = (-1) + 1$ $m = 3$	K1 K1 – find $f(-1)$ K1 – find $\lim_{x \rightarrow -1^-} f(x)$ J1





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

b) ii)	$D_f = (-\infty, -1) \cup (1, +\infty) \cap x \geq 0$ $= x \geq 1$ Vertical asymptote: $\lim_{x \rightarrow 1^+} \frac{3x+1}{\sqrt{x^2-1}} = +\infty$ $\therefore x = 1$ Horizontal asymptote: $\lim_{x \rightarrow +\infty} \frac{3x+1}{\sqrt{x^2-1}}$ $\lim_{x \rightarrow +\infty} \frac{3x+1}{\sqrt{x^2} \sqrt{1-\frac{1}{x^2}}}$ since $x \rightarrow +\infty, \sqrt{x^2} = x$ $\lim_{x \rightarrow +\infty} \frac{\frac{3x}{x} + \frac{1}{x}}{\frac{x}{x} \sqrt{1-\frac{1}{x^2}}}$ $\lim_{x \rightarrow +\infty} \frac{3 + \frac{1}{x}}{\sqrt{1-\frac{1}{x^2}}}$ $= \frac{3+0}{\sqrt{1-0}}$ $= 3$ $\therefore y = 3$	K1 J1 K1 – for only positive infinity K1 – simplify *penalty given if find limit outside of domain. K1 - substitution J1
TOTAL MARKS		14 marks





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

6. a) If $y = \sin(x^2 + 3)$, show that $x \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3y = 0$. (5 marks)

b) The parametric equations of a curve are given by $x = \cot(t) - \operatorname{cosec}(t)$ and $y = \tan(t) - \sec(t)$ where $0 < t < \frac{1}{2}\pi$.

i) Show that $x - \frac{1}{x} = 2 \cot(t)$ and $y - \frac{1}{y} = 2 \tan(t)$.

Deduce the value of $\left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right)$. (5marks)

ii) Find $\frac{dy}{dx}$ and express your answer in terms of x and y only.

Hence, determine the gradient of the curve at the point $\left(-\frac{1}{2}, -\frac{1}{3}\right)$.

(6marks)

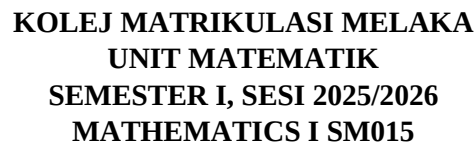




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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

6 b) i)	$ \begin{aligned} x - \frac{1}{x} &= \cot(t) - \operatorname{cosec}(t) - \frac{1}{\cot(t) - \operatorname{cosec}(t)} \\ &= \frac{\cot^2(t) + \operatorname{cosec}^2(t) - 2\cot(t)\operatorname{cosec}(t) - 1}{\cot(t) - \operatorname{cosec}(t)} \\ &= \frac{\cot^2(t) + 1 + \cot^2(t) - 2\cot(t)\operatorname{cosec}(t) - 1}{\cot(t) - \operatorname{cosec}(t)} \\ &= \frac{2\cot^2(t) - 2\cot(t)\operatorname{cosec}(t)}{\cot(t) - \operatorname{cosec}(t)} \\ &= \frac{2\cot(t)(\cot(t) - \operatorname{cosec}(t))}{\cot(t) - \operatorname{cosec}(t)} = 2\cot(t) \text{ shown.} \end{aligned} $ $ \begin{aligned} y - \frac{1}{y} &= \tan(t) - \sec(t) - \frac{1}{\tan(t) - \sec(t)} \\ &= \frac{\tan^2(t) + \sec^2(t) - 2\sec(t)\tan(t) - 1}{\tan(t) - \sec(t)} \\ &= \frac{\tan^2(t) + 1 + \tan^2(t) - 2\sec(t)\tan(t) - 1}{\tan(t) - \sec(t)} \\ &= \frac{2\tan^2(t) - 2\sec(t)\tan(t)}{\tan(t) - \sec(t)} \\ &= \frac{2\tan(t)(\tan(t) - \sec(t))}{\tan(t) - \sec(t)} = 2\tan(t) \end{aligned} $ $ \begin{aligned} \left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right) &= 2\cot(t)2\tan(t) \\ &= 4 \end{aligned} $	K1 – equate denominator for both K1 – use identities to simplify for both J1 J1 –show both J1
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Prepared by :
Mathematics Unit





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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

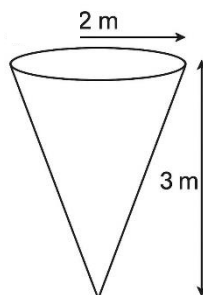
<u>ALTERNATIVES</u>	<u>ALTERNATIVE</u>
$\left(x - \frac{1}{x}\right)\left(1 - \frac{1}{y}\right) = 4$ $\frac{d}{dx}\left[\left(x - \frac{1}{x}\right)\left(1 - \frac{1}{y}\right)\right] = 4$ $\left(1 + \frac{1}{x^2}\right)\left(y - \frac{1}{y}\right) + \left(x - \frac{1}{x}\right)\left(\frac{dy}{dx}\right)\left(1 + \frac{1}{y^2}\right) = 0$ $\left(x - \frac{1}{x}\right)\left(\frac{dy}{dx}\right)\left(1 + \frac{1}{y^2}\right) = -\left(1 + \frac{1}{x^2}\right)\left(y - \frac{1}{y}\right)$ $\frac{(x^2 - 1)(y^2 + 1)}{xy^2} \frac{dy}{dx} = -\frac{(x^2 + 1)(y^2 - 1)}{x^2 y}$ $\frac{dy}{dx} = -\frac{(x^2 + 1)(y^2 - 1)}{x^2 y} \cdot \frac{xy^2}{(x^2 - 1)(y^2 + 1)}$ $\frac{dy}{dx} = -\frac{y(x^2 + 1)(y^2 - 1)}{x(x^2 - 1)(y^2 + 1)}$ at $\left(-\frac{1}{2}, -\frac{1}{3}\right)$ $\frac{dy}{dx} = -\frac{-\frac{1}{3}\left(\left(-\frac{1}{2}\right)^2 + 1\right)\left(\left(-\frac{1}{3}\right)^2 - 1\right)}{-\frac{1}{2}\left(\left(-\frac{1}{2}\right)^2 - 1\right)\left(\left(-\frac{1}{3}\right)^2 + 1\right)} = -\frac{8}{9}$	<u>S</u> K1J1 – correctly differentiate using product rule K1 – simplify J1 K1J1
TOTAL MARKS	16 marks





KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

7. An inverted conical tank with radius $2m$ and height $3m$ is completely filled with water. Water is leaking from the vertex of the tank at a rate of $0.1m^3$ per minute.



At the instant when the height of the water is $1m$, calculate the rate of change of

a) the water level.

[7marks]

b) the radius of the water surface.

[2marks]

NO	ANSWER SCHEME	MARKS
7 a)	$\frac{r}{2} = \frac{h}{3}$ $r = \frac{2h}{3}$ $V = \frac{1}{3}\pi r^2 h$ $= \frac{1}{3}\pi \left(\frac{2h}{3}\right)^2 h$ $= \frac{4}{27}\pi h^3$ $\frac{dV}{dh} = \frac{4}{9}\pi h^2$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \frac{9}{4\pi h^2} \times (-0.1)$ $= -\frac{9}{40\pi h^2}$ at $h = 1$ $\frac{dh}{dt} = -\frac{9}{40\pi(1)^2} = -\frac{9}{40\pi} \text{ m/min}$	K1 – make r as subject K1 – find V in terms of h only. K1 – differentiate correctly K1 – chain rule J1 K1 J1





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UNIT MATEMATIK
SEMESTER I, SESI 2025/2026
MATHEMATICS I SM015

7 b)	$\frac{r}{2} = \frac{h}{3}$ $r = \frac{2h}{3}$ $\frac{dr}{dt} = \frac{2}{3} \frac{dh}{dt}$ $= \frac{2}{3} \left(-\frac{9}{40\pi} \right)$ $= -\frac{3}{20\pi} \text{ m/min}$	K1 – differentiate implicitly. J1
	TOTAL MARKS	9 marks

